

PORTOFOLIO

STUDI KASUS

Prediksi Kehilangan Ketebalan Pipa Menggunakan Random Forest untuk Mendukung Pemeliharaan Prediktif di Sektor Minyak dan Gas

1

PEMANGGILAN DATA

```
Drive already mounted at /content/drive; to attempt to forcibly re-mount
Pipe_Size_mm Thickness_mm Material Grade \
0 800 15.48 Carbon Steel ASTM A333 Grade 6
1 800 22.00 PVC ASTM A106 Grade B
2 400 12.05 Carbon Steel API 5L X52
3 1500 38.72 Carbon Steel API 5L X42
4 1500 24.32 HDPE API 5L X65

Max_Pressure_psi Temperature_C Corrosion_Impact_Percent \
0 300 84.9 16.04
1 150 14.1 7.38
2 2500 0.6 2.12
3 1500 52.7 5.58
4 1500 11.7 12.29

Thickness_Loss_mm Material_Loss_Percent Time_Years Condition
0 4.91 31.72 2 Moderate
1 7.32 33.27 4 Critical
2 6.32 52.45 7 Critical
3 6.20 16.01 19 Critical
4 8.58 35.28 20 Critical
```

2

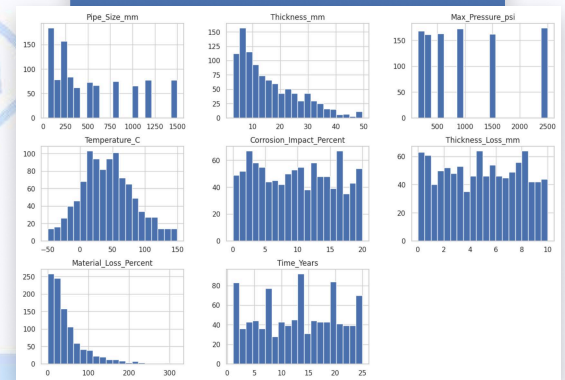
PRA-PEMROSESAN DATA

```
Missing values by column:
Pipe_Size_mm 0
Thickness_mm 0
Material 0
Grade 0
Max_Pressure_psi 0
Temperature_C 0
Corrosion_Impact_Percent 0
Thickness_Loss_mm 0
Material_Loss_Percent 0
Time_Years 0
Condition 0
dtype: int64

Data Types:
Pipe_Size_mm int64
Thickness_mm float64
Material object
Grade object
Max_Pressure_psi int64
Temperature_C float64
Corrosion_Impact_Percent float64
Thickness_Loss_mm float64
Material_Loss_Percent float64
Time_Years int64
Condition object
dtype: object
```

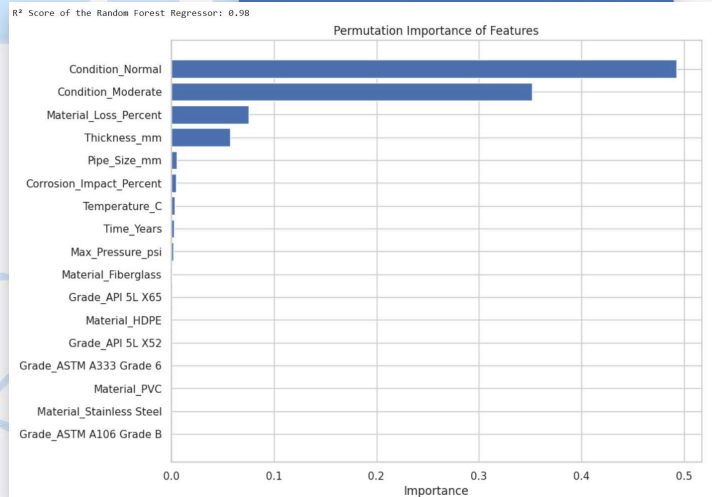
3

ANALISIS DATA EKSPLORATIF



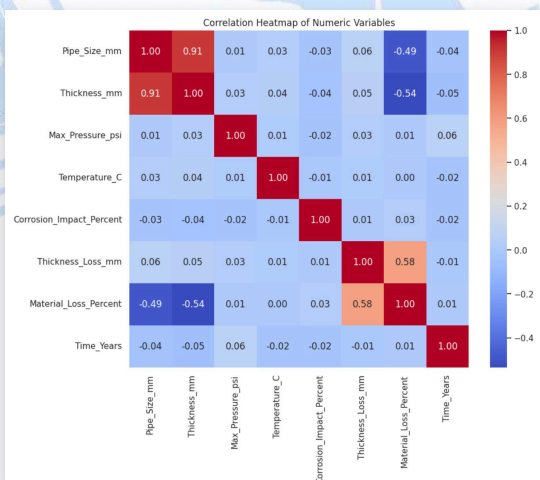
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EVALUASI PREDIKTOR



4

PETA KORELASI



6

VISUALISASI INTERAKTIF

Thickness Loss vs. Time by Condition

